

Customer

Customer Industries Inc

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New York12345  
US

Manufacturer

Steel Testing Labs GmbH

123 Test Street  
10115Berlin  
DE

Digital Material Passport

|            |                   |                  |       |
|------------|-------------------|------------------|-------|
| ID         | TEST-SPECIMEN-001 | Version          | 0.1.0 |
| Issue Date | 2025-09-29        | Certificate Type |       |

Business Transaction

|          |              |             |              |
|----------|--------------|-------------|--------------|
| Order    |              | Delivery    |              |
| Order ID | ORD-2025-001 | Delivery ID | DEL-2025-001 |
| Quantity |              | Quantity    | 1000 kg      |

Product Information

|              |                           |
|--------------|---------------------------|
| Product Name | High Strength Steel Plate |
| Batch ID     | BATCH-2025-001            |

Product Shape

|           |         |
|-----------|---------|
| Form      | Plate   |
| Length    | 6000 mm |
| Width     | 2000 mm |
| Thickness | 50 mm   |

Chemical Analysis

|                 |                   |
|-----------------|-------------------|
| Heat Number     | H123456           |
| Melting Process | EAF+LF            |
| Casting Date    | 2025-09-28        |
| Casting Method  | ContinuousCasting |

Elements

| Symbol | C    | Mn   | Si   | P     | S     | Cr   |
|--------|------|------|------|-------|-------|------|
| Unit   | %    | %    | %    | %     | %     | %    |
| Min    | -    | 1.4  | -    | -     | -     | -    |
| Max    | 0.2  | 1.7  | 0.5  | 0.025 | 0.015 | 0.3  |
| Actual | 0.18 | 1.45 | 0.35 | 0.012 | 0.008 | 0.25 |
| Symbol | Ni   | Mo   | Cu   | Al    | N     | CEV  |
| Unit   | %    | %    | %    | %     | %     | %    |
| Min    | -    | -    | -    | 0.02  | -     | -    |
| Max    | 0.2  | 0.1  | 0.25 | 0.05  | 0.012 | 0.45 |
| Actual | 0.15 | 0.08 | 0.18 | 0.025 | 0.008 | 0.42 |

Formula Definitions

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%

Mechanical Properties

| Property                                                                  | Symbol | Actual | Minimum | Maximum | Method   | Status |
|---------------------------------------------------------------------------|--------|--------|---------|---------|----------|--------|
| <b>Tensile Strength</b><br>Room temperature<br>Specimen: 1/4T, L          | Rm     | 485MPa | 450MPa  | 600MPa  | ASTM E8  | ✓      |
| <b>Charpy V-Notch Impact</b><br>Temperature: -40°C<br>Specimen: 1/4T, L-T |        |        |         |         | ASTM E23 | ✓      |

| Individual Values | # 1 | # 2 | # 3 |
|-------------------|-----|-----|-----|
| Value [ J ]       | 48  | 52  | 45  |

| Statistics | Mean | Min/Max | Std Dev           |
|------------|------|---------|-------------------|
|            | 48.3 | 45 / 52 | 3.5<br>( Sample ) |

|                                                                                   |          |   |
|-----------------------------------------------------------------------------------|----------|---|
| <b>Through-Thickness Tensile</b><br>Room temperature, through-thickness direction | EN 10164 | ✓ |
|-----------------------------------------------------------------------------------|----------|---|

|                     |     |      |     |      |     |
|---------------------|-----|------|-----|------|-----|
| Location (position) | 0   | 0.25 | 0.5 | 0.75 | 1   |
| Value [MPa]         | 475 | 485  | 490 | 487  | 478 |

|                                                                                                             |    |        |        |         |   |
|-------------------------------------------------------------------------------------------------------------|----|--------|--------|---------|---|
| <b>Yield Strength</b><br>Room temperature<br>Specimen: Surface, T - ID: YS-SURF-T-001                       | Re | 355MPa | 335MPa | ASTM E8 | ✓ |
| <b>Elongation</b><br>Room temperature<br>Specimen: Custom (Mid-radius at end section), C - ID: ELONG-MR-001 | A  | 22%    | 20%    | ASTM E8 | ✓ |

Validation

We certify that the material described herein has been tested and inspected in accordance with the specified standards and meets all requirements.

Validated By

| Name             | Title           | Department | Date       |
|------------------|-----------------|------------|------------|
| Dr. Hans Schmidt | Quality Manager |            | 2025-09-29 |